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Percutaneous Aortic Valve Intervention in Patients Scheduled for Noncardiac Surgery: A Japanese Multicenter Study

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ABSTRACT

Background: The optimal management of preexisting severe aortic stenosis (AS) in patients undergoing noncardiac surgery (non-CS) remains uncertain. This study aimed to investigate the safety and effectiveness of percutaneous aortic valve intervention (PAVI) in patients with AS before non-CS.
Methods: We analyzed pooled data within a multicenter Japanese registry from 118 patients with severe AS who underwent PAVI before non-CS. Sixty patients underwent percutaneous balloon aortic valvuloplasty (BAV) and 58 patients underwent transcatheter aortic valve replacement (TAVR). The groups' baseline characteristics, perioperative complications, and 30-day mortality and midterm mortality after non-CS were compared.
Results: The postprocedural mean pressure gradient was higher in the BAV group than in the TAVR group (35.0 ± 11.5 mmHg vs. 11.5 ± 4.8 mmHg, $p < 0.001$). The non-CS operation risk did not differ between the groups ($p = 0.69$). One patient in each group experienced a noncardiac death ($p = 0.74$), and the 30-day mortality rate after non-CS was 1.7%. Heart failure occurred in 2 patients in each group ($p = 0.68$). One patient in the TAVR group experienced a non-disabling stroke, and no myocardial infarctions occurred. Consequently, the combined adverse events were 5.0% and 6.9% in the 2 groups ($p = 0.48$). The bleeding rates during the non-CS were similar in both groups (33.3% vs. 25.9%, $p = 0.25$). There were no differences between the groups regarding midterm mortality ($p = 0.60$), whereas 53.3% of the patients in the BAV group required invasive treatment of their AS during follow-up.
Conclusions: Among patients with severe AS, PAVI before non-CS reduces the AS severity and may contribute to procedural safety during non-CS.

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1. Introduction

Preexisting cardiac disease is a potential source of increased mortality and adverse events during noncardiac surgery (non-CS), and aortic stenosis (AS) is the most common valvular heart disease [1–3]. The classical guidelines recommend surgical aortic valve replacement (SAVR)

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before non-CS in patients with severe AS [1–3]. Recently, some reports have advocated the relatively lower mortality rates associated with more contemporary surgical methods after non-CS in patients with AS, but the rates of major adverse cardiovascular events (MACE) were higher in the AS groups compared with those in the non-AS groups [4–6]. Whereas SAVR is traditionally considered an invasive therapeutic approach for severe AS, transcatheter aortic valve replacement (TAVR) may be a suitable alternative therapeutic approach for high-risk patients with AS or those who are inoperable c to SAVR [7,8]. Percutaneous balloon aortic valvuloplasty (BAV) is another palliative method that reduces AS severity, and the acute success rates associated with BAV have improved as the procedural complications have declined [9]. However, until recently, data have not been available that describe percutaneous aortic valve intervention (PAVI) before non-CS. Therefore, the aim of this study was to investigate the procedural results of PAVI in patients with AS who were scheduled to undergo non-CS.

2. Methods

2.1. Patient population and study design

We analyzed pooled data from the Optimized CathEter vAlvular iNtervention (OCEAN) registry [10,11] that were collected retrospectively between October 2013 and November 2017 from 14 institutions in Japan. BAV data were also collected during the same period. The study's inclusion criteria were patients whose echocardiographic findings showed severe AS and those who were scheduled to undergo elective non-CS within 6 months of undergoing PAVI. The definition of severe AS with normal flow and low flow was previously reported in our registry data [12]. Elective PAVI procedures were performed to reduce the AS severity before non-CS. Especially in patients with asymptomatic AS, decision making the PAVI was determined by anesthesiologist's and local heart team recommendation. Whether TAVR or BAV was undertaken was determined by the individual centers or by the operators in the absence of current recommendations. Fig. 1 presents the patient flow chart showing the assignment of the patients to the BAV and TAVR groups. A total of 121 patients with severe AS underwent PAVI before non-CS. One patient was switched from BAV to TAVR because BAV would have been ineffective and the anesthesiologist disagreed that the patient should undergo non-CS. One patient in the BAV group died before undergoing non-CS and 2 patients in the TAVR group died before undergoing non-CS. Finally, non-CS was performed on 60 patients in the BAV group and on 58 patients in the TAVR group. The patients' baseline characteristics, echocardiographic findings, procedural complications during PAVI, perioperative complications during the non-CS, and 30-day- and midterm clinical outcomes following the non-CS were evaluated, and the BAV and TAVR groups were compared with respect to these variables. The future SAVR or TAVR requirement rates were recorded during follow-up, particularly for the patients who underwent BAV. Each hospital's medical ethics committee had provided broad approval of the study conducted by the OCEAN investigators.

2.2. BAV and TAVR procedures

BAV was mainly performed using a retrograde approach via the femoral artery using semi-compliant or noncompliant balloons. The size of balloon was decided on the echocardiography or multidetector computed tomography. Frequent dilatation was usually attempted until the aortic valve's pressure gradient was minimized. The TAVR procedure has been described in detail previously [10,11]. Procedural and other complications during BAV and TAVR were evaluated using the Valve Academic Research Consortium-2 (VARC-2) criteria [13].

2.3. Non-CS risk evaluation and clinical outcomes

The procedural risk associated with the non-CS was categorized as low, which included minor gynecological, minor orthopedic, minor urological, and breast surgery, intermediate, which included major gynecological, major orthopedic, major urological, intraperitoneal, head, and neck surgery, and high, which included esophagectomy, duodenal-pancreatic surgery, liver resection, pneumonectomy, and total cystectomy [2,3]. The American Society of Anesthesiologists (ASA) physical status classification was determined for each patient. The ASA physical status was calculated for the cohort overall, and the BAV and TAVR groups were compared with respect to these parameters. The causes of death were classified into cardiovascular and non-

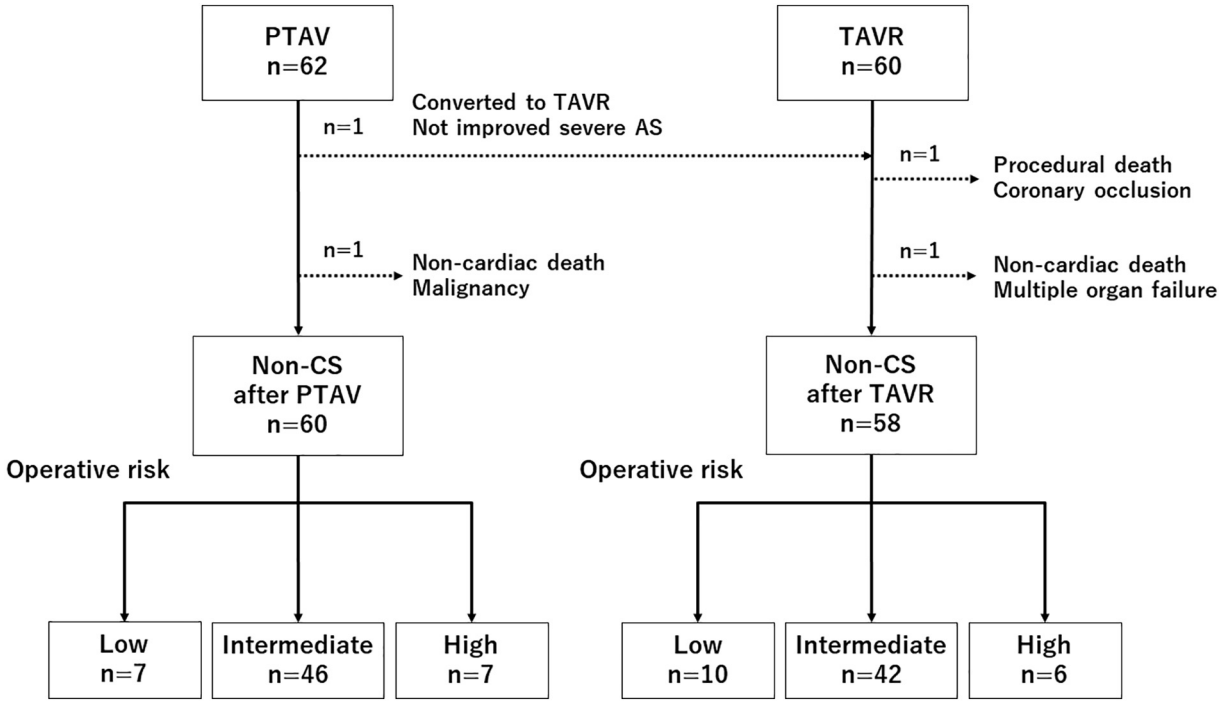


Fig. 1. Study flow chart. The patients were assigned to undergo percutaneous balloon aortic valvuloplasty or transcatheter aortic valve replacement before noncardiac surgery.

cardiovascular deaths. The cardiovascular mortality definition was adopted from the VARC-2 criteria. A perioperative MACE was defined as death, a myocardial infarction (MI), heart failure (HF), or a stroke that occurred after the non-CS. The recent universal definition of MI was applied [14]. HF was defined as an acute decompensated state requiring additional oxygenation, diuretics, vasodilators, other medical therapy, and/or mechanical support. Information about the possible occurrence and/or causes of complications and death was obtained from the hospital charts, or by calling the attending physicians and the patients' families.

2.4. Antiplatelet and anticoagulant therapy during PAVI and non-CS

During PAVI, an intravenous heparin bolus was administered at the start the procedure to achieve an activated clotting time (ACT) of >250 s. There are no strict recommendations for antithrombotic therapy following BAV. In contrast, dual antiplatelet therapy (DAPT) is usually recommended for 6 months, followed by indefinite single antiplatelet therapy (SAPT) after TAVR. The anticoagulant therapy included warfarin as a vitamin K antagonist and a non-vitamin K antagonist. The specific antithrombotic regimen selected was at the discretion of each operator who considered the patient's risk balance between bleeding and thrombotic events before and during the non-CS. Based on each patient's background, we grouped the antithrombotic therapy administered before the non-CS into low-grade antithrombotic therapy that comprised no antithrombotic therapy or SAPT alone, and high-grade antithrombotic therapy that comprised DAPT and/or anticoagulant therapy. Many patients had their antiplatelet and/or anticoagulant therapy stopped to minimize the bleeding complications during the non-CS, and the heparin bridge therapy usage rates were evaluated in these patients. Subgroup analyses compared the bleeding complications, which were evaluated using the VARC-2 criteria, and the transfusion rates during the non-CS.

2.5. Statistical analyses

All of the statistical analyses were performed using IBM®SPSS® software, version 22 (IBM Corporation, Armonk, NY, USA). The continuous variables are expressed as the means and the standard deviations (SDs) or as the medians, depending on the variable's distribution. The categorical data are expressed as percentages of the total. Comparisons of >2 groups were undertaken using chi-squared tests for the categorical covariates, 1-way analyses of variance for the continuous covariates that are expressed as the means and the SDs, and Kruskal-Wallis tests for the continuous variables that are expressed as the medians with the interquartile ranges (IQRs). The Kaplan-Meier method was used to estimate the cumulative mortality rates in the 2 groups. All of the statistical tests were 2-sided, and values of $p < 0.05$ were considered statistically significant.

3. Results

3.1. Baseline patient and procedural characteristics

Table 1 summarizes the patients' baseline characteristics. The frequency of asymptomatic patients was higher in the BAV group (25%) than that in the TAVR group (8.6%) ($p = 0.016$). The prevalence of congestive HF, New York Heart Association classes III/IV, and a history of MI was higher in the TAVR group compared with that in the BAV group. The median Society of Thoracic Surgeons Predicted Risk of Mortality score was lower in the BAV group (4.9 [3.1–7.2]) than that in the TAVR group (5.8 [2.0–8.7]) ($p = 0.05$). Table 2 presents the peri- and postprocedural patient characteristics. The total volume of contrast medium required was greater, and the transfusion, bleeding, and pacemaker implantation rates were higher in the TAVR group than those in the BAV group (all $p < 0.05$). The postprocedural effective orifice

Table 1
Baseline patient characteristics.

	Overall	BAV	TAVR	
Patients, n	$n = 118$	$n = 60$	$n = 58$	p value
Baseline clinical characteristics				
Age, years	82.7 ± 6.1	82.1 ± 6.9	83.3 ± 5.0	0.28
Male, n	40 (33.9%)	19 (31.7%)	21 (36.2%)	0.60
Weight, kg	51.3 ± 11.9	50.3 ± 14.5	52.3 ± 8.6	0.37
Height, cm^2	150.9 ± 9.0	150.3 ± 9.4	151.5 ± 8.7	0.48
BSA, m^2	1.45 ± 0.18	1.43 ± 0.21	1.47 ± 0.15	0.26
BMI, kg/m^2	22.4 ± 4.1	22.1 ± 5.0	22.8 ± 3.1	0.36
Congestive heart failure, n	63 (53.4%)	20 (33.3%)	43 (74.1%)	<0.01
NYHA III/IV, n	41 (34.7%)	16 (26.7%)	25 (43.1%)	0.06
Asymptomatic, n	20 (16.9%)	15 (25.0%)	5 (8.6%)	0.016
Hypertension, n	100 (84.7%)	49 (81.7%)	51 (87.9%)	0.34
Diabetes, n	34 (28.8%)	21 (35.0%)	13 (22.4%)	0.13
Dyslipidemia, n	52 (44.1%)	24 (40.0%)	28 (48.3%)	0.37
Atrial filiation, n	23 (19.5%)	13 (21.7%)	10 (17.2%)	0.54
CKD, n	55 (46.6%)	32 (53.3%)	23 (39.7%)	0.14
Hemodialysis, n	6 (5.0%)	6 (9.8%)	0 (0.0%)	0.01
Peripheral artery disease, n	19 (16.1%)	7 (11.7%)	12 (20.7%)	0.18
COPD, n	12 (10.2%)	6 (10.0%)	6 (10.3%)	0.95
All pulmonary disease, n	18 (15.3%)	8 (13.3%)	10 (17.2%)	0.56
Prior stroke, n	12 (10.2%)	6 (10.0%)	6 (10.3%)	0.95
Prior MI, n	11 (9.3%)	2 (3.3%)	9 (15.5%)	0.02
Cardiac artery disease, n	26 (22.0%)	14 (23.3%)	12 (20.7%)	0.73
Prior PMI, n	2 (1.7%)	0 (0.0%)	2 (3.4%)	0.15
Prior cardiac surgery, n	2 (1.7%)	0 (0.0%)	2 (3.4%)	0.15
Active cancer, n	36 (53.7%)	23 (54.8%)	13 (52.0%)	0.51
Logistic EuroSCORE, %	11.0 (7.4–17.5)	9.7 (5.9–16.4)	11.4 (7.4–17.9)	0.40
EuroSCORE II score, %	3.1 (2.1–4.6)	3.0 (2.2–4.6)	3.3 (2.0–4.7)	0.60
STS score, %	5.6 (3.8–7.8)	4.9 (3.1–7.2)	5.8 (4.3–8.7)	0.05
Laboratory data				
Hemoglobin, g/dl	10.9 ± 2.4	10.7 ± 1.6	11.1 ± 1.9	0.23
BNP, pg/ml	188.3 (77.8–468.9)	184.1 (65.0–484.1)	106.9 (28.9–396.2)	0.64
Serum albumin, g/dl	3.6 ± 0.78	3.6 ± 0.67	3.6 ± 0.59	0.40
Serum creatinine, mg/dl[a]	0.96 ± 0.38	0.95 ± 0.38	0.96 ± 0.34	0.92
eGFR, ml/min/1.73m^2[a]	53.4 (39.0–67.4)	54.6 (33.7–66.2)	53.1 (41.1–69.0)	0.79
Medication				
ACE-I/ARB, n	70 (59.3%)	32 (53.3%)	38 (65.5%)	0.18
Beta blocker, n	34 (28.8%)	16 (26.7%)	18 (31.0%)	0.60
Ca blocker, n	60 (50.8%)	32 (53.3%)	28 (48.3%)	0.58
Digitalis, n	5 (4.2%)	4 (6.7%)	1 (1.7%)	0.18
Diuretic, n	59 (50.0%)	27 (45.0%)	32 (55.2%)	0.27
Tolvaptan, n	13 (11.0%)	3 (5.0%)	10 (17.2%)	0.03
Steroid, n	3 (2.5%)	0 (0.0%)	3 (5.2%)	0.07
Pre-echocardiographic data				
Aortic valve area, cm^2	0.66 ± 0.18	0.68 ± 0.20	0.64 ± 0.14	0.30
Index aortic valve area, cm^2/m^2	0.46 ± 0.12	0.47 ± 0.13	0.44 ± 0.10	0.20
Peak velocity, m/s	4.6 ± 0.76	4.6 ± 0.67	4.5 ± 0.64	0.76
Mean gradient, mmHg	49.3 ± 16.7	49.2 ± 16.5	49.4 ± 16.1	0.96
Peak gradient, mmHg	85.3 ± 24.7	85.7 ± 24.0	84.8 ± 25.9	0.84
Left ventricle ejection fraction, %	58.0 ± 10.6	57.0 ± 9.6	59.1 ± 11.6	0.29
MR grade 3/4, n	8 (6.8%)	3 (5.0%)	5 (8.6%)	0.43
AR grade 3/4, n	9 (7.6%)	3 (5.0%)	6 (10.3%)	0.27

Values are numbers (%), mean ± SD, or median (interquartile range). BAV, balloon aortic valvuloplasty; TAVR, transcatheter aortic valve replacement; BSA, body surface area; NYHA, New York Heart Association; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; MI, myocardial infarction; PMI, pacemaker implantation; EuroSCORE, European System for Cardiac Operative Risk Evaluation; STS score, Society of Thoracic Surgeons Predictive Risk of Mortality; BNP, brain natriuretic peptide; eGFR, estimated glomerular filtration rate; ACE-I, angiotensin converting enzyme inhibitor; ARB, angiotensin II receptor blocker; MR, mitral regurgitation; AR, aortic regurgitation.

[a] Hemodialysis patients excluding.

Table 2
Peri- and postprocedural patient characteristics.

Patients, n	Overall n = 118	BAV n = 60	TAVR n = 58	*p* value
BAV/TAVR procedural variables				
Total procedure time, min	68.5 (49.3–90.0)	64.5 (31.5–90.0)	63.0 (49.0–84.5)	0.42
Contrast volume, ml	90.0 (49.0–120.0)	70.0 (35.0–108.8)	99.5 (59.0–124.3)	0.03
Valve type				
Sapien-XT, n			21 (36.2%)	
Sapien-3, n			25 (43.1%)	
CoreValve, n			4 (6.9%)	
Evolut R, n			7 (12.1%)	
Approach route				
Transfemoral, n			52 (89.7%)	
[a]Alternative approaches, n			6 (10.3%)	
Post-aortic intervention variables				
In hospital stay after procedure, day	10.0 (6.0–25.0)	7.0 (3.3–26.0)	11.5 (7.0–24.3)	0.09
Intensive care unit stay, day	1.0 (1.0–2.0)	1.0 (1.0–2.0)	1.0 (1.0–2.0)	0.37
All stroke, n	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.99
Peri-procedural MI, n	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.99
All bleeding, n	18 (15.4%)	5 (8.5%)	13 (22.4%)	0.04
More than major bleeding, n	14 (11.9%)	5 (8.3%)	9 (15.5%)	0.27
Transfusion, n	23 (19.5%)	7 (11.7%)	16 (27.6%)	0.04
Acute kidney injury, n	2 (1.7%)	0 (0.0%)	2 (3.4%)	0.24
All vascular complication, n	5 (4.2%)	2 (3.3%)	3 (5.2%)	0.68
Major vascular complication, n	3 (2.6%)	1 (1.7%)	2 (3.5%)	0.61
New PMI in hospital, n	4/116 (3.4%)	0/60 (0.0%)	4/56 (7.1%)	0.05
Post-aortic intervention echocardiographic data				
Effective orifice area, cm^2	1.20 ± 0.53	0.83 ± 0.22	1.57 ± 0.37	<0.001
Index effective orifice area, cm^2/m^2	0.83 ± 0.34	0.58 ± 0.14	1.07 ± 0.26	<0.001
Peak velocity, m/s	3.1 ± 1.07	3.9 ± 0.60	2.3 ± 0.45	<0.001
Mean gradient, mmHg	23.4 ± 14.9	35.0 ± 11.5	11.5 ± 4.8	<0.001
Peak gradient, mmHg	42.9 ± 25.7	62.7 ± 18.9	22.0 ± 8.4	<0.001
Not improved severe AS	16 (13.6%)	16 (26.7%)	0 (0%)	<0.001
Left ventricle ejection fraction, %	61.7 ± 14.1	60.8 ± 9.5	62.6 ± 11.0	0.36
MR grade 3/4, n	8 (6.8%)	4 (6.7%)	4 (6.9%)	0.62
AR grade 3/4, n	5 (4.2%)	3 (5.0%)	2 (3.4%)	0.52

Values are numbers (%), mean ± SD, median (interquartile range). AS, aortic stenosis; Other abbreviation as in Table 1.
[a] Alternative approaches included transapical, transiliac, transsubclavian, and direct aortic approach.

area was lower and peak velocity, mean gradient, and peak gradient were significantly higher in the BAV group than those in the TAVR group (all $p < 0.05$). The AS remained severe after BAV in 26.7% of the patients, despite the pressure gradient and mean gradient decreasing. All of the severe AS improved in the patients in the TAVR group.

3.2. Peri- and post-surgical patient characteristics

Table 3 presents the patients' peri- and postsurgical characteristics. Fifteen types of surgery were undertaken in this study. There were no differences between the BAV and the TAVR groups regarding the numbers of patients who underwent low- (11.7% vs. 17.2%), intermediate- (76.7% vs. 72.4%), and high- (11.7% vs. 10.3%) risk non-CS ($p = 0.69$). All of the patients' ASA physical status classifications were >2. More patients were prescribed high-grade antithrombotic therapy after TAVR (53%) than after BAV (25%) ($p = 0.001$), and 20.5% of the patients received heparin bridging therapy. The median times from PAVI, BAV, and TAVR to non-CS were 17.5 days (IQR: 8.0–38.0 days), 11.0 days (IQR: 4.0–18.0 days), and 28.0 days (IQR: 16.8–54.5 days), respectively, ($p < 0.001$).

3.3. Clinical outcomes after non-CS

Fig. 2A and B present the 30-day outcomes after non-CS. The BAV and TAVR groups did not differ regarding the incidence of death within 30 days of the non-CS (1.7% [1/60] vs. 1.7% [1/58], $p = 0.74$). The deaths were caused by acute respiratory failure and peritonitis caused by a duodenal perforation. A non-disabling stroke event occurred in the TAVR group (1.7%). HF developed in 2 patients in the BAV and TAVR groups (3.3% [2/60] vs. 3.4% [2/58], $p = 0.68$). The MACE rates did not differ between the BAV and TAVR groups (5.0% [3/60] vs. 6.9% [4/58], $p = 0.48$). These outcomes were evaluated according to the non-CS risk level (Fig. 2B), and no significant differences were found among the low-, intermediate-, and high-risk groups in relation to the rates of 30-day death ($p = 0.19$), stroke ($p = 0.84$), HF and MI ($p = 0.49$), and MACE ($p = 0.53$). The BAV and TAVR groups did not differ in relation to mid-term mortality after non-CS (Fig. 3). Of the patients in the BAV group, 53.3% (33/60) underwent additional AS treatment involving TAVR (23/60) or SAVR (10/60) during the median follow-up duration of 181 days (78.5–350 days).

3.4. Subgroup analysis of the bleeding complications during non-CS

The overall bleeding complication rates did not differ between the BAV and TAVR groups (33.3% [20/60] vs. 25.9% [15/58], $p = 0.24$), among the low-, intermediate-, and high-operative risk groups (17.6% [3/17] vs. 31.8% [28/88] vs. 30.8% [4/13], $p = 0.50$), between the low-grade and high-grade preprocedural antithrombotic drug groups (33.3% [24/72] vs. 23.9% [11/46], $p = 0.19$), or between those who did and did not use heparin bridge therapy (29.5% [28/95] vs. 30.4% [7/23], $p = 0.56$) (Fig. 4A). The major bleeding rates did not differ between the BAV and TAVR groups (13.3% vs. 5.2%, $p = 0.21$), the low-grade and high-grade preprocedural antithrombotic drug groups (12.5% vs. 4.3%, $p = 0.12$), or between those who did and did not use heparin bridge therapy (10.5% vs. 4.3%, $p = 0.33$) (Fig. 4B). The major bleeding rates tended to be higher in association with the higher surgical risk levels (low: 0% vs. intermediate: 9.1% vs. high: 23.1%, $p = 0.097$) (Fig. 4B).

4. Discussion

The optimal management of preexisting severe AS in patients undergoing non-CS remains uncertain. We believe that our study makes a significant contribution to the literature, because its findings indicate that among patients with severe AS, both strategies between BAV and TAVR before non-CS reduces the AS severity and achieved low mortality during non-CS.

4.1. Non-CS prognosis following BAV or TAVR

The overall 30-day mortality rates after the non-CS were only 1.7% in each group; these deaths were caused by acute respiratory failure and peritonitis caused by a duodenal perforation, and they did not have a cardiac origin. The 30-day mortality rate after non-CS in AS patients reached approximately 10%, but it has improved more recently, and it ranges from 2% to 5%, because of improvements in prescreening, anesthetic techniques, and postoperative care [4–6]. These mortality rates have been stratified according to several factors, including the clinical symptoms associated with the AS, degree of AS severity, ASA physical status, procedure's surgical complexity, and the nature of the non-CS, that is, emergent or non-emergent. [4–6,15,16]. Emergent non-CS was a predictor of 30-day mortality after non-CS [4]. The 30-day mortality rates for AS patients who underwent elective and emergent non-CS were 3.8% and 21.4%, respectively [5]. No cases of emergent non-CS

Table 3
Peri-and postsurgical patient characteristics.

	Overall	BAV	TAVI	
Patients, n	n = 118	n = 60	n = 58	*p* value
Non-CS variables				
Type of surgery				
Anorectal	5 (4.2%)	3 (5.0%)	2 (3.4%)	
Brain	1 (0.8%)	0 (0%)	1 (1.7%)	
Breast	7 (5.9%)	4 (6.7%)	3 (5.2%)	
Foregut	12 (10.2%)	7 (11.7%)	5 (8.6%)	
Hepato-pancreatobiliary	5 (4.2%)	1 (1.7%)	4 (6.9%)	
Gallbladder	5 (4.2%)	4 (6.7%)	1 (1.7%)	
Hip	27 (22.9%)	16 (26.7%)	11 (19.0%)	
Intestinal	30 (25.4%)	17 (28.3%)	13 (22.4%)	0.41
Orthopedic	6 (5.1%)	1 (1.7%)	5 (8.6%)	
Peripheral endoatherectomy/intervention	7 (5.9%)	1 (1.7%)	6 (10.3%)	
Renal	3 (2.5%)	1 (1.7%)	2 (3.4%)	
Thoracic	4 (3.4%)	2 (3.3%)	2 (3.4%)	
Thyroid	1 (0.8%)	0 (0%)	1 (1.7%)	
Urologic	2 (1.7%)	1 (1.7%)	1 (1.7%)	
Vascular	3 (2.5%)	2 (3.3%)	1 (1.7%)	
Surgical operation risk				0.69
Low risk, n	17 (14.4%)	7 (11.7%)	10 (17.2%)	
Intermediate risk, n	88 (74.6%)	46 (76.7%)	42 (72.4%)	
High risk, n	13 (11.0%)	7 (11.7%)	6 (10.3%)	
ASA physical status 2	73 (61.9%)	39 (65.0%)	34 (58.6%)	
ASA physical status 3	40 (33.9%)	20 (33.3%)	20 (34.5%)	0.35
ASA physical status 4	5 (1.7%)	1 (1.7%)	4 (6.9%)	
Low grade antithrombotic therapy	72 (61.0%)	45 (75.0%)	27 (46.6%)	0.001
High grade antithrombotic therapy	46 (39.0%)	15 (25.0%)	31 (53.4%)	
Bridging antithrombotic therapy with heparin, n	23 (20.5%)	9 (15.8%)	14 (25.5%)	0.25
Time to non-CS after BAV/TAVR, day	17.5 (8.0–38.0)	11.0 (4.0–18.0)	28.0 (16.8–54.5)	<0.001
Local anesthesia, n	10 (8.5%)	3 (5.0%)	7 (12.3%)	0.20
Total procedure time, min	130.0 (78.0–205.5)	125.0 (63.3–228.8)	102.0 (27.8–149.3)	0.41
Hospital stay after non-cardiac surgery	18.0 (11.0–26.0)	16.0 (10.0–23.0)	20.0 (10.5–31.0)	0.21
Procedural complication during non-cardiac surgery				
Total bleeding volume, ml	35.0 (0.0–115.0)	25.0 (0.0–110.0)	28.5 (0.0–92.5)	0.75
Rates of transfusion during procedure, n	27 (24.5%)	18 (31.0%)	9 (17.3%)	0.12
Infection in surgical wound after procedure, n	6 (5.1%)	2 (3.3%)	4 (6.9%)	0.32

Values are numbers (%), mean ± SD, or median (interquartile range). Non-CS, non-cardiac surgery; ASA, American Society of Anesthesiologists; CRCI, Revised Cardiac Risk Index; Other abbreviation as in Table 1.

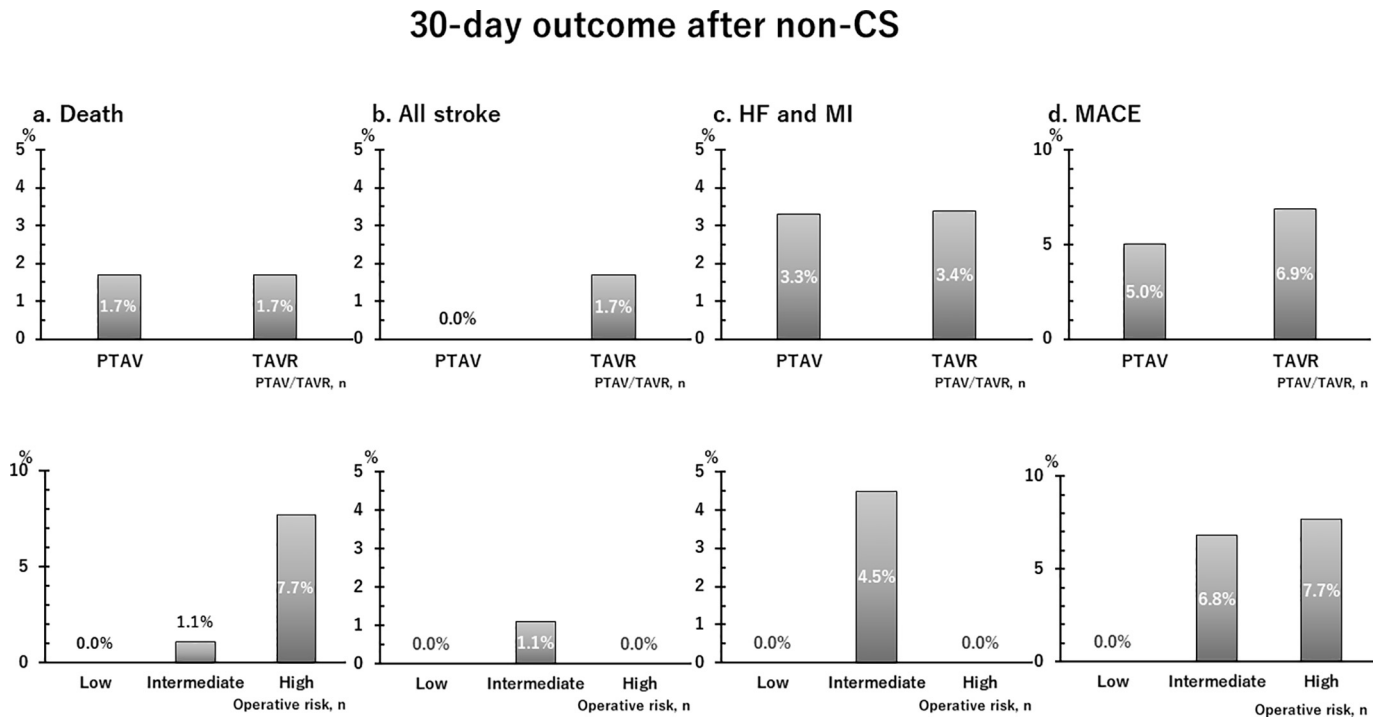


Fig. 2. The 30-day outcome rates after noncardiac surgery (CS). (A) The 30-day outcome rates for death, stroke, heart failure and myocardial infarction, and major adverse cardiac event in the percutaneous balloon aortic valvuloplasty and transcatheter aortic valve replacement groups after non-cardiac surgery. (B) The 30-day outcome rates for death, heart failure and myocardial infarction, stroke, and major adverse cardiovascular events in the low, intermediate, and high-risk non-cardiac surgery groups.

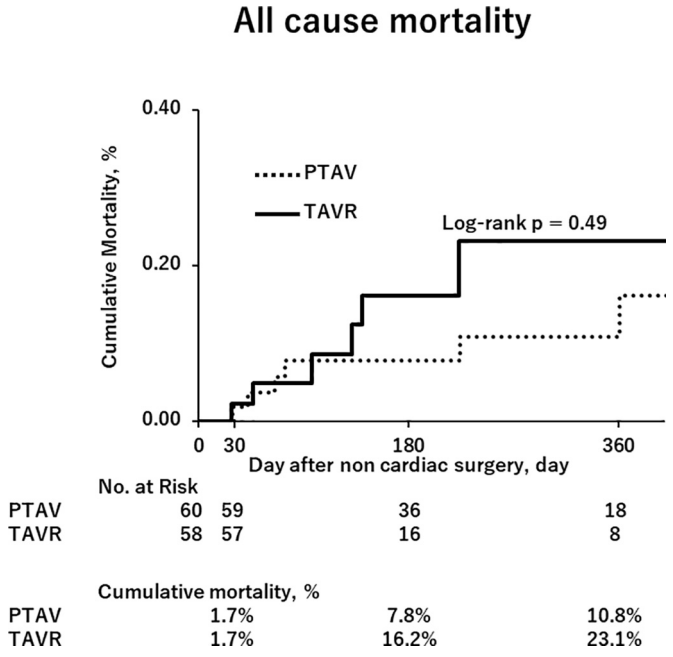


Fig. 3. All-cause mortality in the percutaneous balloon aortic valvuloplasty (BAV) and transcatheter aortic valve replacement (TAVR) groups after noncardiac surgery (CS). The BAV and TAVR groups did not differ regarding midterm mortality after non-CS.

were included in this study; therefore, the effect of AS treatment on emergent non-CS cannot be described. However, excluding emergent non-CS, the aforementioned factors were not associated with an increased risk of mortality following non-CS in this study. A reduction in the 30-day mortality rate following PAVI for AS was broadly achieved. Notably, we minimized the effects of the heterogeneous backgrounds of the patients with severe AS who underwent non-CS.

4.2. MACE rates for non-CS following BAV or TAVR

Multiple comorbidities are strongly correlated with a higher incidence of MACE in patients with AS compared with those without AS before non-CS [4–6,15,16]. The patients' baseline clinical characteristics were similar to or more advanced than those reported from previous studies [4–6,15,16]. Nonetheless, the overall MACE rate was 5.9% with relatively fewer procedural complications after the non-CS, regardless of the presence of multiple comorbidities. A previous report also revealed that patients with symptoms attributable to AS had a higher incidence of MACE compared to patients without symptoms [17]. In our study, there were no mortalities and only a minor stroke without any neurological damage at discharge occurred in the asymptomatic AS sub-group. The rates of complications were significantly low in the weak symptom group, supporting the previous finding.

The current guidelines for the management of patients with severe AS who are undergoing non-CS recommend assessing the clinical symptoms first, then evaluating the level of risk associated with the non-CS and SAVR [1–3,7,8]. Because the patients posed a high risk for SAVR or were inoperable with respect to alternative procedures, the choice of procedure was limited to BAV or TAVR; however, this study's results may provide new insights with respect to the potential benefits of less invasive BAV or TAVR.

4.3. Bleeding complications during non-CS

Bleeding complications during invasive treatments are another important clinical concern. The bleeding complication rates during non-CS were positively correlated with the level of surgical risk, but not with the antithrombotic regime, heparin bridge therapy usage, or the choice of BAV or TAVR. If a robust strategy was established regarding the discontinuation of SAPT/DAPT and heparin bridge therapy before non-CS, it would minimize the bleeding complications that occur during non-CS. Obviously, more invasive and complex non-CS poses a greater

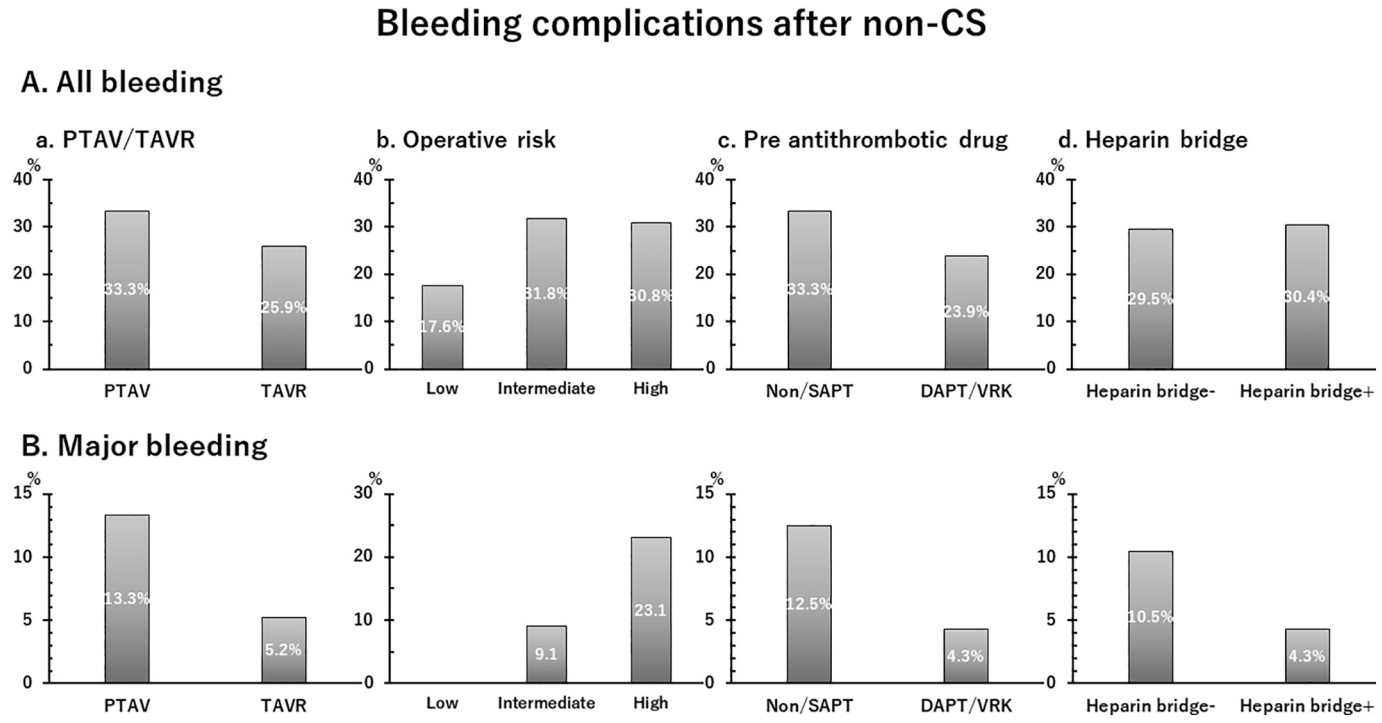


Fig. 4. Bleeding complications after noncardiac surgery (CS). (A) All bleeding events and (B) major bleeding events in the patients in the percutaneous balloon aortic valvuloplasty and transcatheter aortic valve replacement, the low-, intermediate-, and high-risk operative risk grade, the low-grade and high-grade preprocedural antithrombotic drug operative, and use and nonuse of heparin bridge therapy subgroups.

risk of bleeding [1–8]; hence, general surgeons should continue to seek the best way to prevent periprocedural bleeding. Although no definitive recommendation exists that addresses antithrombotic treatment before non-CS, the differences in preprocedural prescriptions may not be a concern if clinical decisions adequately account for each patient's background.

4.4. BAV and TAVR characteristics

BAV and TAVR have advantages and disadvantages. BAV is a simple, low cost procedure that does not require a strict antithrombotic therapy regimen before non-CS; however, its ability to stabilize AS is limited. Of the patients with severe AS, 26.7% of the patients were classified as having severe AS after BAV, despite declines in the peak velocity and gradient. Consequently, 1 patient required TAVR before they underwent non-CS. During the follow-up period, TAVR or SAVR was performed after the non-CS on 58.8% of the patients who underwent BAV. These reasons were mainly explained by restenosis after BAV, worsening of clinical symptom or occurrence of HF. In contrast, TAVR is a radical therapy for severe AS. Surgeons and anesthesiologists feel more comfortable during non-CS when the patient's severe AS is completely cured. However, the devices used for TAVR are expensive, and patients generally require continuous SAPT/DAPT after the procedure. Nevertheless, a recent report has suggested that DAPT has a higher incidence of bleeding and similar rates of death, MI, and stroke compared to SAPT alone after TAVR [18]. There is data supporting the short duration of DAPT or SAPT alone after TAVR. In addition, the antithrombotic agents were not associated with an increased risk of bleeding during non-CS. A life expectancy of <1 year is a contraindication for TAVR [1]; thus, palliative therapy, including BAV, should be selected for these patients. In other words, TAVR may be beneficial for reducing the procedural risk and future intervention for AS. Anyway, we cannot focus on the treatment differences in BAV and TAVR because of the several selection bias. Both strategies should be determined clinically based on the individual patient's background. As such, this issue was beyond our study scope.

4.5. Limitations

This study's results should be interpreted in the context of its limitations. First, the lack of data concerning the control group or historical cohort in patients with severe AS without PAVI before non-CS is clearly a limitation. Second, the study's participants were not randomized to undergo BAV or TAVR, and the individual centers and operators determined whether patients underwent BAV or TAVR. The actual number of untreated patients who did not undergo non-CS because they were deemed to be even too high risk for BAV or TAVR also is lacking. Hence, a significant selection bias was inevitable. Third, the small sample volume of this study was underpowered to reveal statistically significant differences; thus the similar clinical results for BAV and TAVR cannot be overstated. Moreover, the degree of AS after BAV varies as still severe, moderate, and mild. Although the treatment effect of BAV should be assessed, this also is limited because of the small sample volume. Further large-scale investigations will be required to clarify the optimal strategy and management in patients with AS undergoing non-CS. Finally, the evaluation of the occurrence of stroke was not confirmed by neurological specialists or precise magnetic resonance imaging assessments. None of the patients experienced a disabling stroke after the non-CS, and all of the patients, except those who died, were discharged without new-onset neurological dysfunction.

4.6. Conclusions

Although we should focus on the balance between the risks and benefits of invasive treatments, the results from the current study, which showed excellent early mortality rates and low MACE rates, may support more aggressive interventions for patients with severe AS to reduce the perioperative risks before they undergo non-CS.

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Disclosures

Dr. Yamamoto, Dr. Tada, Dr. Naganuma, Dr. Shirai, Dr. Mizutani, and Dr. Watanabe are clinical proctors for Edwards Lifesciences and Medtronic. Dr. Araki, Dr. Tabata, Dr. Takagi, Dr. Higashimori, and Dr. Hayashida are clinical proctors of Edwards Lifesciences. Dr. Ueno is a clinical proctor for Medtronic. The remaining authors have nothing to disclose.

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